

Yuze Fu

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 Website |  LinkedIn |  Github

RESEARCH INTERESTS

Numerical Methods for Partial Differential Equations, Physics-based Computer Graphics

EDUCATION

- University of Toronto Scarborough** Sep 2023 - June 2027 (expected)
Honours Bachelor of Science Scarborough, Ontario, Canada
 - GPA: 3.99/4.00
 - Double Majored in Mathematics and Computer Science
 - Math Courses: Abstract Algebra, Numerical Methods for PDEs, Number Theory, Graph Theory, Analysis
 - CS Courses: Machine Learning, Operating System, Database, Human-computer Interaction
- High School Affiliated to Fudan University** Sep 2020 - June 2023
High School Diploma Shanghai, China
 - Studied Math, Physics, Chemistry, Geography, Chinese, English
 - Member of the school's competitive programming team for the Informatics Olympiad
 - Member of the school's soccer team, playing against other high schools in Shanghai

HONORS AND AWARDS

- Department of Computer Science Award** 2026
University of Toronto \$12000
 - 16-Weeks Undergraduate Student Research Award funded by the Department of Computer Science.
- University of Toronto Scholar** 2024
University of Toronto \$1500
 - University-wide in-course scholarship recognized for most outstanding academic achievement during the 2023-2024 session.
- Dean's List** 2024, 2025
University of Toronto Scarborough
- Second Prize in the National Olympiad in Informatics (NOIP2020)** 2020
China Computer Federation

RESEARCH EXPERIENCE

- Adaptive Particle Splitting for Fluid Simulation** May 2026 - Present
Supervised by Prof. Eitan Grinspun and Jonathan Panuelos Dynamic Graphics Project Lab
- Automated Misinformation Detection & Claim Verification Pipeline** Jun 2025 - Aug 2025
Research Assistant Rotman School of Management
 - Developed a two-phase NLP pipeline using SBERT and LLMs to classify social media misinformation.
 - Applied cosine similarity and embedding-based filtering to align thousands of tweet-article pairs.
 - Leveraged DeepSeek API to classify claim relationships as identical, contradictory, or unrelated.
 - Optimized system accuracy via iterative threshold tuning and incremental data processing strategies.

TEACHING EXPERIENCE

- Teaching Assistant for CSCB09 (System Programming)** Winter 2026
- Teaching Assistant for CSCB36 (Introduction to the Theory of Computation)** Fall 2025
- Teaching Assistant for MATA22 (Linear Algebra I for Mathematical Science)** Winter 2025
- Teaching Assistant for CSCA67 (Discrete Mathematics)** Fall 2024

PROJECTS

• **Classification of Finitely Generated Modules over PIDs**

Mar 2026 - Apr 2026

Abstract Algebra Essay

- Classified finitely generated modules over PIDs to unify the structure of abelian groups and operators.
- Formulated the bijection between $F[x]$ -modules and linear maps to prove operator structural identities.
- Constructed Rational and Jordan Canonical Forms using Invariant Factor and Elementary Divisor forms.
- Concluded with a review of the theoretical and computational trade-offs between different matrix canonical forms.

• **Numerical Analysis of Finite Difference Schemes for Hyperbolic PDEs**

Feb 2026 - Apr 2026

Numerical Methods for PDEs Projects

- Implemented 8 finite difference schemes using vectorized NumPy operations, optimizing performance for advection-based simulations.
- Evaluated the stability and order of convergence for standard schemes (e.g. Lax-Wendroff, Leapfrog) under varying CFL conditions.
- Developed SBP-SAT boundary treatments for high-order Runge-Kutta methods, ensuring stable and consistent convergence up to the 4th order.
- Analyzed theoretical vs. experimental trade-offs between ghost cell extrapolation and SBP-SAT operators for boundary stability.

• **NSI Forecasting: Machine-Learning-Driven Neighborhood Safety Index**

Nov 2025 - Dec 2025

Machine Learning Project

- Engineered a custom Neighborhood Safety Index (NSI) based on the Canadian Crime Severity Index to quantify urban safety levels.
- Evaluated Ridge, KNN, and FNN models to compare baseline linear accuracy against non-linear neural approximations.
- Developed an LSTM network achieving a leading R^2 of 0.8295 by modeling complex 12-month temporal sequences.
- Optimized performance through grid search tuning of architecture depth, activation functions, and k-neighbors.

COMMUNITY INVOLVEMENT

• **UTSC Accessibility Services**

Sep 2024 - Present

Volunteer Notetaker

- Produced comprehensive lecture and tutorial notes to support students with diverse accessibility needs.
- Courses: Probability (STAB52), Computation Theory (CSCB36), Group Theory (MATC01), Numerical Methods for PDEs (MATD10)

• **UTSC Summer School**

Jul 2025 - Aug 2025

Teaching Assistant

- Helped Professors to hold seminars about computer science topics for high school students.
- Answered students' questions and provided information about university lives.

• **The Association of Mathematical and Computer Science Student (AMACSS)**

Sep 2024 - Apr 2025

Course Representative

- Developed review materials and facilitated pre-exam seminars to improve peer academic performance.
- Courses: Discrete Math (CSCA67), Introduction to Computer Science (CSCA08)

• **UTSC Google Developer Group (GDG)**

Oct 2024 - Apr 2025

Academic Director

- Supported the execution of "Build with AI," an industry-focused event with 500+ registrations and live AI demos.
- Created an interactive emotion-matching game that successfully increased participant engagement.

SKILLS AND HOBBIES

- **Programming Languages:** C, Python, Assembly, Java, R, LaTeX, Markdown
- **Hobbies:** Soccer, Piano, Swimming, Traveling, Movie, RPG Game